

AI LEARNING & STUDY ASSISTANT

INTERNSHIP PROJECT REPORT

1. ABSTRACT

The AI Learning & Study Assistant is an AI-agent based educational application designed to help students learn from course materials. The system uses Ollama as the local language model, Python for development, VS Code as the development environment, Retrieval-Augmented Generation (RAG) for searching uploaded learning materials, ChromaDB for vector storage, and Streamlit for the user interface. The assistant can answer questions from uploaded PDF materials, generate study plans, create quizzes, and maintain basic student memory.

2. INTRODUCTION

Students often spend considerable time searching notes and preparing revision schedules. This project provides a single AI-based assistant that can interact with course materials and support common study activities. The application combines an AI agent with RAG, memory, and tools through a simple chat interface.

3. OBJECTIVES

- Develop an AI Learning & Study Assistant using Python.
- Use Ollama for local AI model interaction.
- Implement RAG for uploaded PDF course materials.
- Provide a study-plan generation tool.
- Provide a quiz/MCQ generation tool.
- Maintain basic student information and conversation memory.
- Develop and execute the project inside a Python virtual environment using VS Code.

4. SOFTWARE AND TECHNOLOGIES USED

- Python 3.12 – application and AI-agent development.
- Ollama – local language model execution.
- Visual Studio Code – coding and project execution.
- Streamlit – interactive web interface.
- ChromaDB – vector database for retrieval.
- Sentence Transformers – text embeddings.
- PyPDF – PDF text extraction.
- JSON – simple student-memory storage.

5. SYSTEM ARCHITECTURE

Student → Streamlit Interface → AI Learning Agent → Tool Selection → RAG / Study Plan / Quiz → Ollama → Final Response

For RAG, the uploaded PDF is converted into text, split into chunks, embedded into vectors, stored in ChromaDB, and searched using the student's question. Retrieved course context is supplied to the local language model to produce a grounded response.

6. PROJECT MODULES

6.1 User Interface

The Streamlit interface provides a chat area for questions, PDF upload functionality, and access to study tools.

6.2 AI Agent

The LearningAgent manages interaction and routes requests to course-material search, study-plan generation, quiz generation, or general question answering.

6.3 RAG Module

The KnowledgeBase extracts PDF text, creates chunks, generates embeddings, stores them in ChromaDB, and retrieves relevant chunks for a query.

6.4 Memory Module

Student name, study goal, and recent conversation history are stored in a JSON file so the assistant can use previous interaction information.

6.5 Study Plan Tool

The tool creates a seven-day plan containing study, revision, and practice activities.

6.6 Quiz Tool

The quiz tool uses a topic and retrieved course context to generate five MCQs with four options, the correct answer, and a short explanation.

7. PROJECT FOLDER STRUCTURE

AI_Learning_Study_Assistant/

- |— venv/
- |— app.py
- |— agent.py
- |— rag.py

```
|— requirements.txt
|— memory.json
|— README.md
|— .gitignore
|— chroma_db/
```

8. IMPLEMENTATION

The project is implemented in Python and executed inside a virtual environment. Required packages are installed using requirements.txt. Ollama provides the local model and Streamlit starts the application.

Create environment: `python -m venv venv`

Activate on Windows: `venv\Scripts\activate`

Install packages: `pip install -r requirements.txt`

Run application: `streamlit run app.py`

9. WORKING PROCEDURE

- Student opens the application.
- Student uploads a course PDF.
- RAG extracts and stores document information.
- Student asks a question or requests a tool.
- AI agent identifies the required operation.
- Relevant chunks are retrieved from ChromaDB for material-based questions.
- Retrieved context is sent to Ollama.
- Answer is displayed in Streamlit.
- Student details and recent conversation history are stored.

10. SAMPLE INPUTS

- What is artificial intelligence?
- Explain this topic from my uploaded material.
- Create a study plan.
- Generate a quiz about machine learning.
- My name is Student.
- My goal is to complete Python this week.

11. EXPECTED OUTPUT

The system provides natural-language answers, generates a seven-day study plan, and produces five MCQs. When course material is available, RAG supplies relevant content to the AI model.

12. ADVANTAGES

- Simple student-friendly interface.
- Uses local AI through Ollama.
- Answers questions using uploaded learning materials.
- Reduces effort in preparing study plans and quizzes.
- Combines AI agent, RAG, memory, and tools.
- Can be extended with additional subjects and tools.

13. LIMITATIONS

- Answer quality depends on the uploaded material and local model.
- Scanned/image-only PDFs may require OCR.
- Memory currently uses a local JSON file.
- Study planning is basic and has no calendar integration.

14. FUTURE ENHANCEMENTS

- Multiple student profiles.
- OCR for scanned PDFs.
- Progress tracking and analytics.
- Voice interaction.
- Calendar integration.
- Flashcards and automatic revision tests.

15. CONCLUSION

The AI Learning & Study Assistant demonstrates how an AI agent can combine a local language model, RAG, memory, and tools to support student learning. It provides a practical educational use case and can be extended into a personalized learning platform.

16. REFERENCES

- Python Documentation.
- Ollama Documentation.
- Streamlit Documentation.
- ChromaDB Documentation.
- PyPDF Documentation.